



Subtopic: Multiprocessor Systems

1. Introduction

- **Definition:** A multiprocessor system is a computer system with **two or more processors (CPUs)** that share a common memory and are interconnected to work together.
 - **Goal:** Improve performance, reliability, and throughput by executing multiple tasks simultaneously.
 - Multiprocessor systems are widely used in **servers, supercomputers, and high-performance computing (HPC)**.
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2. Characteristics of Multiprocessor Systems

1. **Multiple CPUs:** At least two or more processing units.
 2. **Shared Memory (in most systems):** All processors access the same memory space.
 3. **Interconnection Network:** Processors communicate via buses, crossbars, or networks.
 4. **Operating System Support:** Requires special OS (symmetric multiprocessing OS) to manage scheduling, synchronization, and communication.
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3. Types of Multiprocessor Systems

A. Symmetric Multiprocessing (SMP)

These notes were created by Muhammad Abdullah Ahsan, an admin from the GCUF GUIDERS channel. Join here: [GCUF GUIDER](#).



- All processors are **equal peers**.
- Shared memory is accessible by all CPUs.
- OS treats all processors uniformly.
- Example: Modern **multi-core CPUs** (Intel, AMD).

Advantages:

- Simple programming model (shared memory).
- High reliability (failure of one CPU doesn't stop the system).

Disadvantages:

- Memory contention (all CPUs competing for the same memory bus).
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B. Asymmetric Multiprocessing (AMP)

- Processors have **different roles**: one master CPU controls the system, others act as slaves.
- Slave processors execute tasks assigned by the master.

Example: Early embedded systems, some real-time systems.

Advantages:

- Simpler hardware/OS design.

Disadvantages:

- Master processor is a bottleneck.



C. Distributed Memory Multiprocessors (Clusters / MPP)

- Each processor has **its own private memory**.
- Processors communicate via high-speed interconnection networks (message passing).
- Examples: Supercomputers, cloud data centers.

Advantages:

- Scales to thousands of processors.
- Avoids memory contention.

Disadvantages:

- Programming is complex (explicit communication required).
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4. Interconnection Structures in Multiprocessor Systems

- **Bus-Based:** Simple, cost-effective, but not scalable (used in small SMPs).
 - **Crossbar Switch:** Every CPU connected to every memory module (fast but expensive).
 - **Multistage Networks (Omega, Butterfly):** Used in large-scale multiprocessors.
 - **Message Passing Networks:** Used in distributed multiprocessor systems.
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5. Memory Organization in Multiprocessors

1. **Shared Memory:** All CPUs access the same memory (SMP).
 - Requires **cache coherence protocols** (MESI, MOESI).
 2. **Distributed Memory:** Each CPU has private memory; communication via messages.
 3. **Hybrid (NUMA – Non-Uniform Memory Access):** Memory is physically distributed but logically shared.
 - Access to local memory = fast
 - Access to remote memory = slower
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6. Advantages of Multiprocessor Systems

- **Increased Performance:** Multiple tasks executed simultaneously.
 - **Reliability:** Fault-tolerant (if one CPU fails, others continue).
 - **Resource Sharing:** Memory, I/O devices shared.
 - **Scalability:** Additional processors can be added (in some architectures).
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7. Challenges in Multiprocessor Systems

- **Synchronization:** Ensuring tasks coordinate properly (locks, semaphores).
- **Cache Coherence:** Keeping copies of data in caches consistent.
- **Memory Bottleneck:** High contention in shared-memory systems.



- **Complex OS Support:** Requires multiprocessor-aware scheduling.
 - **Programming Difficulty:** Writing parallel programs efficiently is challenging.
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8. Real-World Examples

- **SMP:** Intel Xeon processors in servers.
 - **NUMA:** AMD EPYC processors.
 - **Cluster (Distributed Memory):** Google Cloud, Amazon AWS, IBM supercomputers.
 - **Hybrid Supercomputers:** Combining CPUs + GPUs (e.g., Summit, Fugaku).
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✓ Summary

- Multiprocessor systems contain **multiple CPUs** that work together to improve performance.
- Classified into **SMP, AMP, Distributed Memory, and NUMA** systems.
- Require **efficient interconnection networks, memory organization, and cache coherence protocols**.
- Widely used in **servers, data centers, and supercomputers** but face challenges like synchronization, cache coherence, and programming complexity.